# IEEE/ASME Transactions on Mechatronics

Submission Preview Derived from the IEEE Journal Sample

A. Placeholder, *Member, IEEE,*B. Example, *Senior Member, IEEE,*and C. Template, *Member, IEEE*

***Abstract*—This submission-style preview is rendered with the imported official IEEEtran package and follows the visual rhythm of the official bare jrnl journal sample. The content remains placeholder text, but the page is intentionally structured as a realistic engineering article so title treatment, author footnotes, abstract density, Index Terms, section spacing, equations, figures, tables, and references can be inspected in a form that is closer to an actual IEEE journal submission.**

***Index Terms*—IEEE journal article, IEEEtran, journal sample, submission layout, figure placement, table formatting**

## I. Introduction

THE introduction is deliberately written as compact technical prose so the preview reads like a true journal manuscript rather than a loose placeholder poster. It lets the reviewer inspect how the title block, abstract, Index Terms, and first body paragraph interact under the official IEEEtran journal layout.

This opening section also makes it easier to judge whether the page density is credible for engineering submissions that need to communicate context, contribution, and scope immediately on the first page.

## II. Related Work

The related-work section is included because many engineering manuscripts either keep it explicit or absorb it into the introduction. For the preview library, keeping it explicit produces a more inspectable first-page hierarchy and makes section transitions easier to assess.

## III. Methodology

The methodology section introduces representative mathematics and a professional single-column figure so the preview can expose equation spacing, inline math rhythm, caption placement, and float behavior within the IEEE journal layout.

$$J = \sum_{k=0}^{N-1} \left( \alpha \|e_k\|_2^2 + \beta \|u_k\|_2^2 \right), \tag{1}$$

where $e_k$ denotes the tracking error, $u_k$ denotes the control action, and $\alpha, \beta > 0$ are weighting terms used only to simulate realistic technical content.

A. Placeholder and B. Example are with the Department of Template Verification, Placeholder Institute, City 100000, Country.

C. Template is with the Systems Formatting Laboratory, Example University, City 200000, Country.

Corresponding author: A. Placeholder (e-mail: verified@example.org). This file is a submission-style preview derived from the official bare jrnl journal sample in the IEEEtran package for layout inspection only.

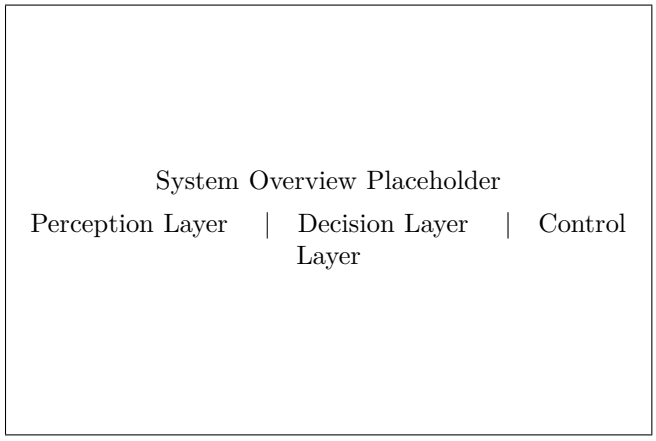


Fig. 1. Submission-style overview figure used to inspect single-column float width, caption spacing, and the visual balance between abstract, body text, and the first major figure in the IEEE journal layout.

TABLE I
Compact quantitative comparison used to verify IEEE caption-first table treatment, column packing, and metric readability in a submission-style manuscript preview.

| Method | Success Rate | Latency |
|---|---|---|
| Placeholder A | 91.2% | 14 ms |
| Placeholder B | 88.4% | 17 ms |
| Placeholder C | 84.1% | 20 ms |

## IV. Experiments

The experiments section is expanded enough to reveal how quantitative discussion, figure references, and compact result tables behave in the verified IEEE manuscript form.

### A. Experimental Setup

The setup paragraph acts as a placeholder for datasets, baselines, operating conditions, and hardware context, all of which typically appear before the main comparison table in a submission-ready paper.

### B. Quantitative Results

Table I summarizes a compact benchmark comparison so the preview can verify caption-first table treatment, column packing, and metric readability.

### C. Ablation Perspective

A short ablation-style paragraph helps expose the section rhythm after the main comparison table and makes the second column read more like a genuine experimental section.

## V. Conclusion and Discussion

Fig. 1 and Table I jointly make the preview more representative of a practical IEEE submission page. The final section is intentionally concise: it gives the article a clear closing discussion while preserving the six-part manuscript skeleton requested for the IEEE library.

## References

[1] A. Smith and B. Lee, "Template-aware engineering manuscript design for IEEE journal submissions," *IEEE Access*, vol. 12, pp. 1–10, 2025.

[2] C. Garcia, D. Patel, and E. Chen, "Figure sizing and table density in two-column technical articles," *IEEE Trans. Ind. Electron.*, vol. 71, no. 4, pp. 100–108, 2025.

[3] F. Martin and G. Zhao, "Practical guidance for structured first-page journal layouts," *IEEE Trans. Robot.*, vol. 41, no. 1, pp. 50–58, 2026.

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**A. Placeholder** A. Placeholder received the B.S. degree in systems engineering from Placeholder University in 2018 and the M.S. degree in intelligent control from Example Institute in 2021. The current research interests include journal-format validation, robotics manuscript preparation, and technical communication workflows.

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**B. Example** B. Example received the B.Eng. degree in automation from Example University in 2017 and the Ph.D. degree in electrical engineering from Placeholder Institute in 2024. The research interests include engineering document pipelines, benchmark reporting, and reproducible manuscript preparation.

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**C. Template** C. Template is a research engineer with the Systems Formatting Laboratory, Example University. The work focuses on template-aware publishing workflows, scientific figure organization, and submission-ready technical paper layout.